

Sanjay Samala

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OBJECTIVE

AI/ML engineer passionate about building scalable multilingual AI systems, retrieval-augmented generation (RAG) pipelines, and OCR datasets for low-resource languages. Experienced in developing production-ready AI applications using LLMs, vector search, and modern backend frameworks.

EDUCATION

Matrusri Engineering College

Bachelor of Engineering in Computer Science and Engineering

Nov. 2022 – May 2026

CGPA: 7.0

EXPERIENCE

AI Developer Intern — Viswam.ai

Oct 2025 – Feb 2026

Low-Resource Language AI Initiative (Meta x Viswam.ai x IIIT Hyderabad)

- Designed metadata extraction pipeline for a Telugu literacy corpus (**758 books, 10,000+ stories, 1947-2012**) using Gemini 2.5 Flash with structured JSON schemas for a story-level indexing.
- Built **Telugu OCR dataset: 120k+ image-text pairs**, 10.7GB, published on Hugging Face.

PROJECTS

- Telugu Agentic RAG System** — *Generative AI, Streamlit, Google Gemini OCR, RAG* [GitHub] [Live Link]
 - Problem:** Telugu AI produces "Translated English" - grammatically passable but culturally hollow.
 - Engineered **multi-agent Planner -& Drafter -& Critic** loop, retrieval over **10,000+ stories** via GTE Multilingual + Qdrant: **92% Hit Rate @1, 99% @5, MRR ~0.95**.
 - Designed **Validator Agent** (rule-based + LLM critique) rejecting drafts with passive voice, lazy descriptions, or cultural inaccuracies - quality enforced architecturally, not by prompting.
 - Stack:** Multi-agent, Qdrant, GTE Multilingual, Gemini/OpenAI/Groq APIs, Streamlit.
- NatyaBhoda – AI Dance Learning System** — *Computer Vision, MediaPipe, XGBoost, Streamlit* [GitHub]
 - Engineered a real-time **gesture recognition pipeline** that utilizes pose estimation to classify Kuchipudi hand mudras.
 - Developed a low-latency Streamlit application that **provides live inference and instant feedback** for AI-assisted dance learning.

TECHNICAL SKILLS

- Languages & Databases:** Python, C, HTML, JavaScript, MySQL
- AI, GenAI & Vision:** Agentic RAG, Multi-Agent Systems (LangGraph/LangChain), Reflexion, LLMs, Dense Vector Search, MediaPipe, OCR, Prompt Engineering
- Frameworks & Libraries:** PyTorch, Hugging Face, Scikit-learn, XGBoost, Pandas, NumPy
- Backend & Infra:** FastAPI, Express.js, React.js, Streamlit, Qdrant, REST APIs, Gemini/OpenAI/Groq APIs Git

ACHIEVEMENTS AND CERTIFICATIONS

- Qualified for the **Final Round of Youth Speak Hackathon at IITHyd**; participated in **4+ Hackathons** and attended **8–10 technical events/workshops**.
- Published a **75K+ Telugu stories dataset** on Hugging Face for AI/LLM and retrieval-based applications.
- Designed a **metadata extraction pipeline** for Telugu literary corpus containing **758 books and 10,000+ stories (1947–2012)** using Gemini 2.5 Flash API with structured JSON schemas.
- Built and published a **Telugu OCR dataset** with **120K+ image-text pairs (10.7GB)** for OCR fine-tuning and document understanding tasks.
- Contributed to **3 production-level AI projects** during internship at **Viswam.ai**, currently deployed in live environments.
- Serving as **Technical Lead** and **Treasurer** of **DevCatalyst Technical Club**, managing technical initiatives, hackathons, and club operations.
- Certifications:** Saylor Academy Python Programming, Udemy Python Programming.